

Baiyi Lian

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SUMMARY

I am an AI Developer with a background in machine learning and linguistics. I build and deploy AI applications such as TTS (Text-to-Speech) systems, voice agents, and language learning tools, with a focus on performance and cost optimization. I am experienced throughout the machine learning lifecycle, from data collection and preparation to model training, fine-tuning, evaluation, and deployment. I adapt to new challenges and aim to develop robust, human-centered AI systems that deliver meaningful real-world impact.

EDUCATION

University of Helsinki, Helsinki, Finland

Master of Arts | Language Technology

Sep. 2023 – Jun. 2025

- GPA: 3.52/5.0
- Relevant Coursework: Deep Learning, Building NLP Application, Models and Algorithms in NLP, Advanced Programming

Shanghai University of Engineering Science, Shanghai, China

Bachelor of Management | Logistics Management(Air Freight)

Sep. 2016 – Jun. 2020

- GPA: 3.7/4.0
- Relevant Coursework: Logistics Management, Marketing, Economics, Mathematics

WORK EXPERIENCE

Language Trampoline Project (KONE foundation), Helsinki, Finland

Consecutive Interpreter

Dec. 2025 – present

- Worked as a freelance English-Chinese interpreter on the Language Trampoline project (KONE Foundation), meeting parents of non-Finnish-native toddlers and preschoolers and collaborating with the PhD researcher to develop language support strategies for their children.
- Facilitated communication between the researcher and families from diverse linguistic and cultural backgrounds, ensuring accurate information exchange and supporting qualitative research activities.
- **SKILLS:** OPI interpretation, Cross-cultural Communication, Active Listening

The Water Bear Strategy Ltd., Helsinki, Finland

AI Developer

May. 2026 – Jun. 2026

- Built a privacy-first local legal AI server MVP for solo-lawyer document workflows.
- Set up the core AI system so legal documents could be searched and used in chat which includes local AI model hosting, a web chat interface, document search/indexing, and secure browser access over HTTPS.
- Created a clean folder structure, added checks to confirm services are running, prepared places for logs/backups, and made sure passwords, private files, and client documents would not be accidentally uploaded to GitHub.
- **SKILLS:** LM Studio, RAG, Python, Open WebUI, Docker, Hetzner Cloud

TRAINING/ACTIVITIES

University of Helsinki, Helsinki, Finland

Event Participant - Machine Translation Marathon 2025

Aug. 2025 – Aug. 2025

- Taken part in the Machine Translation Marathon 2025.

YMCA Tokyo Japanese Language School, Tokyo, Japan

International Student

Apr. 2022 – Apr. 2023

- Attained Japanese Language Proficiency Test (JLPT) at the highest level N1.
- Main courses: Grammar, Vocabulary, Speaking, Writing, Japanese Culture.

PROJECTS

Calm, Personal project

Topic: Multilingual Mental Health Support Voice Agent

Apr. 2026 – present

- Built a compassionate, evidence-based voice AI supportive guide for helping users manage stress, anxiety, fatigue, depression, self-esteem issues, and insomnia.

- Implemented a Direct-to-Model WebSocket strategy, bypassing the need for a traditional back-end to minimize latency.
- Built a custom CI/CD pipeline using GitHub Actions to securely inject API secrets into a Vite production build for Firebase Hosting.
- Designed a reactive 'Orb' interface using Tailwind CSS and Framer Motion to provide visual feedback during active listening and speaking states.
- **SKILLS:** Voice agent, React, CI/CD, Firebase hosting, Tailwind CSS, Typescript, Node.js

LILT, *SkyhouseZhou*

Topic: AI-powered Language Learning Application

Apr. 2026 – Jun. 2026

- Collaborated on a web-based and mobile Finnish learning application that leverages LLMs to provide personalized learning paths and interactive exercises for more than 70 Finnish learners.
- Integrated local LLM for text generation and TTS (Text-to-Speech) for the AI chat feature that supports various daily life scenarios, enhancing users' daily conversational skills.
- Designed evaluation metrics to analyse users' speech patterns and provide real-time feedback on scalable metrics such as pronunciation, intonation, and fluency, helping users improve their spoken Finnish skills effectively.
- **SKILLS:** Full-stack development, local LLM integration, Text-to-Speech, Ollama, ngrok

Easy-TTS-Reader, *Personal Project*

Topic: AI-powered PDF-to-Audio Desktop Application

Dec. 2025 – Jun. 2026

- Built a cross-platform Electron desktop application that converts PDF documents to natural-sounding speech using three TTS engines, local Kokoro-82M, cloud OpenAI TTS, and macOS System TTS with 54 voices across 9 languages.
- Integrated Kokoro-82M, an open-weight TTS model with 82 million parameters, running entirely locally for free speech synthesis with automatic language detection and voice selection, without any API keys or subscriptions.
- Developed a Python-based Kokoro TTS wrapper that spawns as a subprocess from Electron, streaming generated WAV audio chunks back for real-time playback with configurable speed control (0.5x-2.0x).
- Implemented a batch CLI converter for processing multiple PDF files at once using Kokoro, enabling audiobook-style batch conversion with configurable voice and file pattern filters.
- Designed a clean dark-themed UI with PDF drag-and-drop, text extraction via pdf-parse, audio export functionality, and fully local processing guaranteeing document privacy.
- **SKILLS:** Text-to-Speech (TTS), Electron, Node.js, Python, Kokoro-82M, PDF Text Extraction, Desktop Application Development

Aichologist, *Aichologist*

Topic: AI-based support for mental wellbeing

Apr. 2026 – May. 2026

- Contributed to a full-stack AI psychologist chat application with over 80 paid subscriptions of Finnish users.
- The app contains two main components, Python/FastAPI as back-end and Flutter as front-end, deployed on Hetzner Cloud with Docker cross mobile and web platform.
- Implemented a real-time voice feature using Elevenlab STT and TTS API, enabling users to interact with the AI psychologist through voice input for a more natural and accessible experience.
- **SKILLS:** Real-time speech recognition, Elevenlab API, FastAPI, Flutter, Docker, Hetzner Cloud

TeamsCaptionTrans, *Personal Project*

Topic: Real-time Teams Live Caption Translation Browser Extension

Dec. 2025 – Dec. 2025

- Developed a Chrome/Edge browser extension that automatically captures live captions from Microsoft Teams meetings via DOM mutation observers and translates them bidirectionally between Chinese and English in real-time.
- Integrated Google Translate API for automatic language detection, determining whether captions are in Chinese or English and translating to the opposite language without manual switching.
- Built a clean popup interface displaying real-time translations with language direction indicators, source/target language labels, and translation history with timestamps persisted via Chrome Storage API.
- Designed a robust caption detection system using MutationObserver with periodic health-check fallbacks, ensuring reliable caption capture across dynamic Teams UI changes.
- Used Webpack for module bundling, structuring the extension with content scripts for DOM interaction, background scripts for state management, and a dedicated popup for user interface.
- **SKILLS:** Chrome Extension Development, Real-time Speech Translation, Google Translate API, JavaScript, Webpack, DOM Manipulation

CERTIFICATES

EF SET: EF SET Certificate: C2 (Apr. 2026)

Coursera: [Introduction to Psychology](#) (Feb. 2026)
HuggingFace: [Fundamentals of Agents](#) (Dec. 2025)
DataCamp: [Associate AI Engineer for Data Scientists](#) (Dec. 2025)
IBM: [Python for Data Science, AI & Development](#) (Aug. 2023)
Microsoft Certified: [Azure AI Fundamentals](#) (Jun. 2023)

TECHNICAL SKILLS

AI & Machine Learning: Large Language Models (LLMs), RAG Systems, PyTorch, Tensorflow, HuggingFace, Scikit-learn, Reinforcement Learning (Gymnasium, RLHF)
Speech & Voice Technology: Text-to-Speech (TTS), Speech-to-Text / Speech Recognition, Voice Agent, Real-time Speech Translation, Local TTS Models (Kokoro-82M)
Agentic Frameworks: smolagents, LangChain, LlamaIndex
Data Engineering: Snowflake, Supabase, SQL, dbt, MatLab
MLOps & Cloud: Docker, Kubernetes, Azure DevOps, GitHub Actions, CI/CD, Firebase (Hosting/Auth), Hetzner Cloud
Software & Languages: Python, Node.js, TypeScript, JavaScript, React, Tailwind CSS, Electron, HTML/CSS, Chrome Extension Development, Shell Scripting, R, High-Performance Computing (HPC), LaTeX
Infrastructure: Git/GitLab/GitHub, Azure OpenAI Service, LM Studio, Ollama

LANGUAGE SKILLS

Chinese (Mandarin): Native
Cantonese (Cantonese): Native
English: C2 (EF SET Certified)
Japanese: N1 (JLPT Certified)
Finnish: A2

DRIVING LICENSE

Mainland China: C1
Finland: B
Japan: C1 (Valid until 2025-09-22)

REFERENCES

Available upon request.
